Benchmark Study of World-Class Faculties Comparable to the Faculty of Geoscience and Technology

Tsinghua University | China University of Geosciences (Wuhan) | China University of Geosciences (Beijing) | Massachusetts Institute of Technology (MIT) | Colorado School of Mines

Research-and-interpretation report, phase 1 | 12 November 2026

Purpose and scope

This report collects publicly available information on faculties abroad whose academic structure resembles the Faculty of Geoscience and Technology: a 4-year undergraduate programme that combines geology and ore deposits, geophysical and geochemical prospecting, drilling engineering, petroleum, mineral engineering, civil engineering, surveying/geospatial information and space/atmosphere science. For each university it gives the comparable unit, size figures, academic structure (with an organisation diagram), curriculum totals, the semester-by-semester subject list with credits for a representative programme, short syllabus content, and the main laboratories.

How to read this report

- Figures are the latest the universities publish online; the year of each figure is given. Faculty/department-level student counts are often not published - where only university-level numbers exist, that is stated.
- One representative bachelor programme is shown in full per university (the one closest to your Mineral Exploration / Civil Engineering specialisations). Other programmes are listed by name.
- Credit systems differ: Chinese credits are roughly 16 class hours each; US semester hours are 1 lecture hour per week for a semester; MIT counts 'units' (12 units = roughly 4 US semester hours).
- Syllabus content: Colorado School of Mines and MIT publish course descriptions, which are paraphrased here. Tsinghua and CUG publish course lists and credits but not full syllabi online; their 'content summary' column is a short description of the standard content of that subject, written for this report, and is marked with an asterisk (*).
- Laboratory pictures: university image servers block automated downloading, so photographs could not be embedded. Each laboratory entry gives the official web page where photos are published.

Summary: which unit corresponds to which of your institutes

No single foreign faculty reproduces all six of your institutes plus the drilling division. China University of Geosciences (Wuhan and Beijing) is by far the closest overall match, because it was built from the same Soviet-style 'geological university' model in 1952. Colorado School of Mines is the closest Western match for resources, petroleum, mining and geological engineering. Tsinghua and MIT are strong in civil/hydraulic engineering and earth-system/planetary science but have no mining, drilling or ore-deposit units.

Your institute / division	Tsinghua	CUG (Wuhan)	CUG (Beijing)	MIT	Colorado School of Mines
Resource exploration (geology, ore deposits, geochemical prospecting)	None (geology moved out in 1952)	School of Earth Sciences; School of Earth Resources	School of Earth Sciences and Resources	EAPS - Geology, Geochemistry + Geobiology	Dept. of Geology and Geological Engineering
Civil engineering (building, prospecting, tourist district)	School of Civil Engineering (Civil, Hydraulic, Construction Mgmt)	School of Engineering (Civil, Urban Underground Space, Geological Eng.)	School of Engineering and Technology (Civil, Geological Eng.)	Dept. of Civil and Environmental Eng. (separate)	Civil and Environmental Eng. (other college)
Mineral engineering (mineralogy, materials, rock use)	-	School of Materials Science and Chemistry; Gemological Institute	School of Materials Science and Technology; School of Gemology	-	Metallurgical and Materials Eng.; Mining Eng.
Geophysical prospecting (EM, magnetic, equipment)	-	School of Geophysics and Geomatics; Mech. Eng. and Electronic Information	School of Geophysics and Information Technology	EAPS - Geophysics; Earth Resources Laboratory	Dept. of Geophysics
Petroleum (prospecting, mining, machines, processing)	-	School of Earth Resources (Petroleum Engineering)	School of Energy Resources	-	Dept. of Petroleum Engineering
Space (space science, atmosphere, geospatial info)	Dept. of Earth System Science; Hydraulic Eng. (remote sensing)	School of Geography and Information Engineering	School of Land Science and Technology	EAPS - Atmospheres/Oceans/Climate; Planetary Science	Space Resources programme (graduate)
Drill engineering division	-	School of Engineering - Exploration Technology and Engineering	School of Engineering and Technology; deep-drilling key lab	-	Petroleum Eng. (drilling); Mining Eng.

Key figures at a glance

University / unit	Students	Staff / researchers	Bachelor credits	Length
Tsinghua - School of Civil Eng. + Dept. Earth System Science	Univ.-level only; Civil Eng. Dept. has trained ~10,000 undergraduates and ~5,000 postgraduates	Civil Eng. Dept. 71 staff (53 faculty); DESS 30 faculty	163-169 credits	4 years, 8 semesters + 3 summer terms
CUG (Wuhan) - whole university / Faculty of Engineering	33,125 full-time (university); 1,939 undergrad + 1,199 graduate (Faculty of Eng.)	3,303 staff (555 professors); Faculty of Eng. 214 staff, 145 instructors	~170-180 (see CUGB plan)	4 years
CUG (Beijing)	16,758 FTE students (THE data)	16.6 students per staff; 11 CAS + 1 CAE academicians	177 credits (Resources Prospecting Eng.)	4 years, 8 semesters + summers
MIT - EAPS (Course 12)	175 graduate students; 25 undergraduate majors; 60 postdocs	40+ faculty	GIRs + 180-198 units (major 144-150 units)	4 years
Colorado School of Mines	8,247 (6,371 UG / 1,876 grad, fall 2025)	603 academic staff	130 semester hours (Geological Eng.)	4 years + 5-week field session

1. Tsinghua University (Beijing, China)

1.1 Comparable units

Tsinghua has no geology or mining faculty: its geology department was transferred out in 1952 to help found the Beijing Institute of Geology, which became China University of Geosciences. The comparable units today are the School of Civil Engineering (civil, hydraulic/geotechnical engineering, construction management, surveying) and the Department of Earth System Science (atmosphere, global change, earth-system big data and remote sensing). Together they cover your Institute of Civil Engineering and part of your Laboratory of Space.

1.2 Size: researchers, students and graduates

Item	Figure / detail	Source
Dept. of Civil Engineering staff	71 staff, of whom 53 are faculty	civil.tsinghua.edu.cn (General Introduction)
School of Civil Engineering research base	3 national first-class key disciplines; 17 research institutes and 9 laboratories	civil.tsinghua.edu.cn (History of SCE)
Dept. of Civil Engineering research units	7 research institutions, 1 national engineering lab, 1 MoE key lab, 1 Beijing engineering centre	civil.tsinghua.edu.cn
Graduates (historical)	The civil engineering programme has trained nearly 10,000 undergraduates and almost 5,000 postgraduates	SCE 2021 brochure
Hydraulic Eng. undergraduate intake	Approx. 90 students per year	civil.tsinghua.edu.cn (Undergraduate)
Dept. of Earth System Science faculty	30 faculty: 2 academicians, 15 professors, 10 associate, 3 assistant professors	dess.tsinghua.edu.cn (Introduction)
DESS degrees	PhD in Ecology; Master in Atmospheric Science; new 4-year BSc 'Earth System Science' (major code 070604T); minor of 12 courses / 30 credits since 2022	dess.tsinghua.edu.cn
Ranking	QS Civil and Structural Engineering: No. 9 / 10 / 7 (2019 / 2020 / 2021)	yz.tsinghua.edu.cn

Note: Tsinghua does not publish current headcounts of students per department on its English pages; the figures above are what is published.

1.3 Academic system

Students are admitted into a broad 'Civil, Hydraulic and Ocean Engineering' category. Years 1-2 are common (university general education, mathematics/physics, and a shared engineering platform); from the summer term after Year 2 each student chooses one of five directions. Each year has an autumn semester, a spring semester and a short summer term used for internships and field practice. The final semester is a 15-credit thesis.

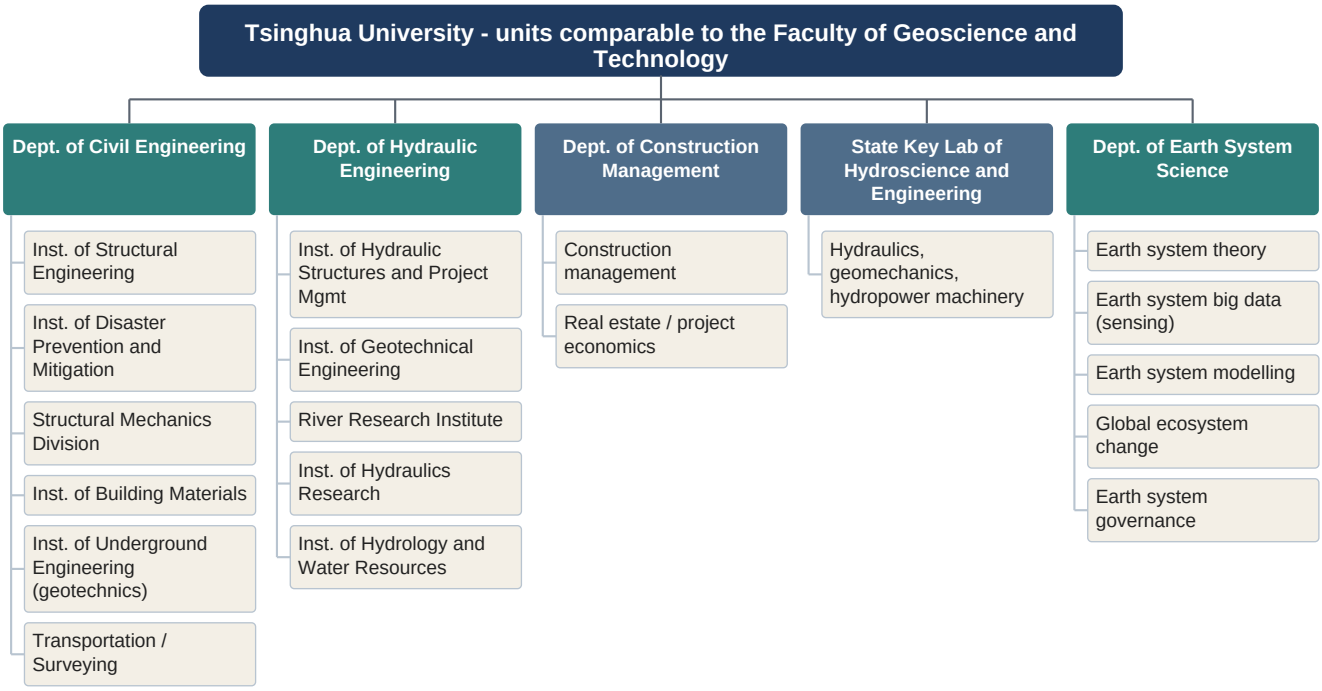


Diagram drawn for this report from the departments' published structure. Teal = closest match to your institutes.

1.4 Curriculum: Civil, Hydraulic and Ocean Engineering (2023 plan)

Block	Credits	Content
University general education	47	Political theory 17, PE 4, foreign language 8, writing and communication 2, general electives 11, military training 4 (+ swimming test)
Mathematics and natural science	33	Calculus A(1)(2) 10, linear algebra 4, probability and statistics 3, maths elective 3, physics 4 + physics lab 2, computing 5, chemistry/biology/earth science 2
Category platform	34	Category core 22 (mechanics, fluids, materials, engineering geology, surveying, economics, project management); development electives 6; experiments/practice 6
Direction (major) courses	44-50	Civil Eng. 50, Hydraulic Science and Eng. 50, Transportation 48, Ocean 47, Construction Mgmt 44 - incl. summer practice and 15-credit thesis
Category extension	6	Courses from another direction
Total	163-169	4 years: 8 semesters + 3 summer terms; Bachelor of Engineering

1.5 Directions (courses) offered

- Civil Engineering
- Hydraulic Science and Engineering
- Transportation Engineering
- Ocean Science and Engineering
- Construction (Engineering) Management
- Earth System Science (separate BSc, Dept. of Earth System Science)

1.6 Subjects by semester with credits - Civil Engineering direction

*Content summaries marked * are short descriptions of the standard content of each subject, written for this report.*

Semester	Code	Subject	Cr.	Content summary
Y1 Autumn	20030181	Intro to Civil, Hydraulic and Ocean Eng.	1	Overview of the disciplines, careers and landmark projects*
	10421324	Linear Algebra	4	Matrices, vector spaces, eigenvalues*
	10421055	Calculus A(1)	5	Limits, differentiation, integration of one variable*
	20120152	Fundamentals of Engineering Graphics	2	Projection, sections, engineering drawing standards*
	30250023	Computer Language and Programming	3	Programming fundamentals (computing elective)*
	10680053	Ethics and Rule of Law	3	Required political course
	10691342	Writing and Communication	2	Academic writing and oral communication
	-	English (1); PE (1); Situation and Policy (1)	2+1+1	University requirements
Y1 Spring	10421065	Calculus A(2)	5	Multivariable calculus, series*
	10431174	University Physics CE	4	Mechanics, electromagnetism, waves*
	20900012	Engineering Computer Drafting	2	CAD drafting practice*
	10440012	University Chemistry B	2	Chemistry for engineers (or Intro Biology)*
	20740112	Data Structures and Algorithms	2	Computing elective*
	10610193	Outline of Modern Chinese History	3	Required political course
	-	English (2); PE (2); freshman seminars	2+1+1-2	University requirements
Y1 Summer	40030711	Cognition (field-orientation) Practice	1	1.5 weeks of site visits to structures and works*
Y2 Autumn	20310334	Theoretical Mechanics	4	Statics, kinematics, dynamics of rigid bodies*
	10420803	Probability and Statistics	3	Distributions, estimation, testing*
	10430782	Physics Experiments A(1)	2	Laboratory physics*
	30030352	Engineering Economics	2	Time value of money, project appraisal*
	40030352	Building Materials	2	Cement, concrete, steel, asphalt properties*
	40030361	Building Materials Lab	1	Testing of construction materials*
	-	Principles of Marxism; English (3); PE (3)	3+2+1	University requirements
Y2 Spring	20310343	Mechanics of Materials	3	Stress, strain, bending, torsion, buckling*
	30040493	Fluid Mechanics	3	Hydrostatics, flow equations, pipe and channel flow*
	40040152	Engineering Geology	2	Rocks, soils, structures and geohazards for engineering sites*
	30910052	Engineering Project Management I	2	Planning, scheduling, cost and quality control*
	-	Surveying	1	Levelling, angle and distance measurement*
	-	Mao Zedong Thought...; English (4); PE (4)	2+2+1	University requirements
Y2 Summer	40030402	Surveying Practice	2	2-week field surveying camp*
	30030622	Mathematical Modelling and Data Analysis	2	Modelling and data analysis for engineering*
	-	Political practice	2	Social practice
Y3 Autumn	20030044	Structural Mechanics (1)	4	Statically (in)determinate structures, force and displacement methods*
	30040513	Soil Mechanics	3	Soil properties, consolidation, shear strength, earth pressure*
	30030522	Building Architecture	2	Building design principles and construction details*
	40031102	Fundamentals of Architecture	2	Architectural design basics*
	-	Category / general electives	~6	Chosen from elective lists

Semester	Code	Subject	Cr.	Content summary
Y3 Spring	20030153	Concrete Structures 1	3	Reinforced concrete members design*
	30030113	Steel Structures (1)	3	Steel members and connections*
	30040162	Foundation Engineering	2	Shallow/deep foundations, piles*
	30030323	Civil Engineering CAD	3	Computer-aided structural design*
	30910032	Building Construction Technology	2	Construction methods and equipment*
	30030601	Earthquake Engineering and Dynamics	1	Seismic response basics*
Y3 Summer	40030584	Construction Internship	4	5 weeks on a construction site*
Y4 Autumn	40031184	Integrated Course Design (Civil)	4	Team design project of a structure*
	-	Electives	~4	Elective courses
Y4 Spring	40030700	Comprehensive Thesis Training	15	Bachelor thesis

1.7 Hydraulic Science and Engineering direction (years 2-4, differences only)

Semester	Code	Subject	Credits
Y2 Summer	20030191	Surveying Practice	1
	30040551	Geology Field Practice	1
	30030622	Math Modelling and Data Analysis	2
Y3 Autumn	30040513	Soil Mechanics	3
	30040533	Structural Analysis	3
	30040573	Principles and Applications of Hydrology	3
	20040143	River Hydraulics	3
	30040142	Rock Mechanics (elective)	2
Y3 Spring	30030583	Reinforced Concrete	3
	40040753	Hydraulic Structures	3
	30040263	Foundation Engineering (elective)	2
Y3 Summer	40041003	Production Internship	3
Y4 Autumn	40040964	Hydraulic Engineering Design	4
	40040942	Water Resources Planning and Management	2
Y4 Spring	40040590	Comprehensive Thesis Training	15

Selected electives relevant to your faculty: Rock Mechanics, Underground Cavern Engineering, Urban Geotechnical Engineering, Remote Sensing Principles and Methods, Fundamentals of Geospatial Information Technology, Marine Resources and Energy, Marine Geology, Offshore Petroleum Engineering, Ocean Geotechnical Engineering.

1.8 Laboratories

Laboratory	What it is	Photos (official page)
Key Laboratory of Civil Engineering Safety and Durability (Ministry of Education)	Founded 1985; 4,400 m2 with 3 structural testing halls and a construction materials laboratory; integrates the structural, disaster-prevention, mechanics, materials and geotechnical institutes	civil.tsinghua.edu.cn/ceen/Research/Research_Institutions_and_laboratories.htm
State Key Laboratory of Hydroscience and Engineering	National lab for hydraulics, river dynamics, geomechanics/geotechnics of dams and caverns, and hydropower machinery; rated 'Best Laboratory' in 2008 and 2013 national assessments	sklhse.tsinghua.edu.cn
Institute for Underground Engineering - geotechnical lab	Soil and rock mechanics, soil dynamics, numerical and centrifuge modelling	civil.tsinghua.edu.cn/ceen/info/1052/1341.htm
Old and New Hydraulic Buildings	Historic hydraulic experiment hall (1933) and New Hydraulic Building (1955), physical river and dam models	civil.tsinghua.edu.cn/heen
DESS 'Tsinghua DESS' satellite and observation platforms	Department's own satellite launched 11 Nov 2024 for earth observation; earth-system modelling and big-data facilities	dess.tsinghua.edu.cn/en/news

Sources for this section:

Tsinghua 2023 undergraduate training plan, Civil, Hydraulic and Ocean Engineering: tsinghua.edu.cn/jxjywj/bkzy2023/zxzy/10.pdf

Dept. of Civil Engineering: civil.tsinghua.edu.cn/ceen/About/General_Introduction.htm; History of SCE: civil.tsinghua.edu.cn/en/About/History_of_SCE.htm
Dept. of Earth System Science: dess.tsinghua.edu.cn/en/About/Introduction.htm; new major: dess.tsinghua.edu.cn/en/info/1222/2404.htm
Graduate admissions overview: yz.tsinghua.edu.cn/en/info/1014/1057.htm

2. China University of Geosciences (Wuhan) - CUG

2.1 Comparable units

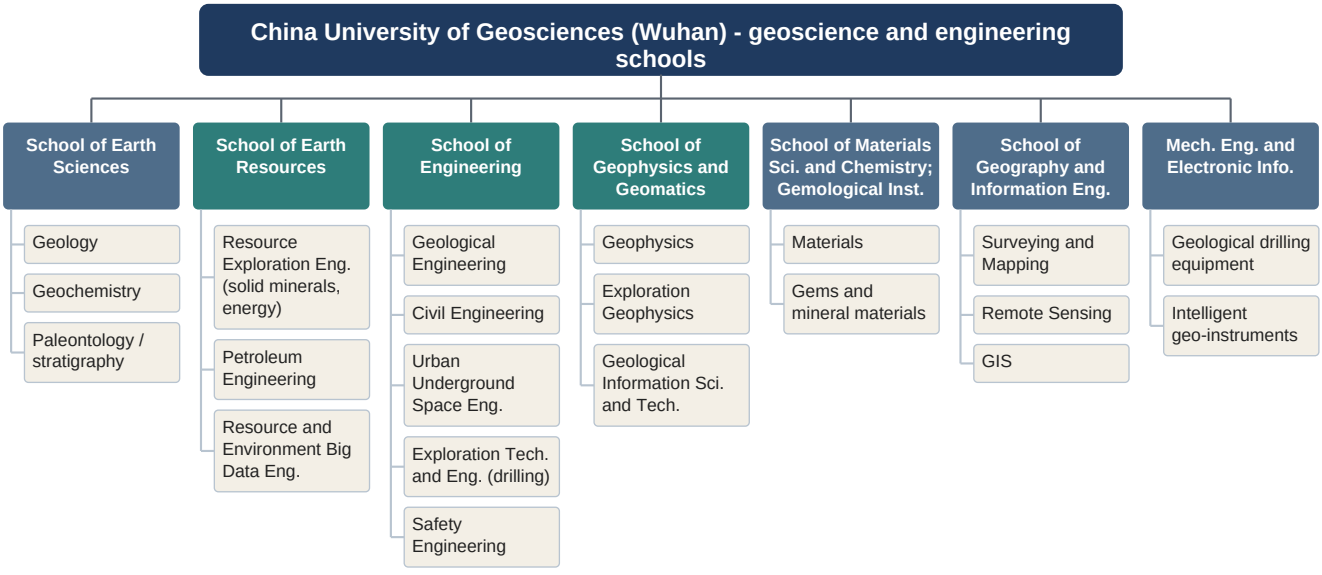
CUG is a whole university organised around the same objects as your faculty. Its predecessor, the Beijing Institute of Geology (1952), was formed from the geology departments of Peking, Tsinghua, Tianjin and other universities. Its Geology and Geological Resources and Engineering disciplines are ranked first in China. The schools listed below together correspond almost one-to-one with your six institutes; the Faculty (School) of Engineering is the single closest unit because it combines civil engineering, geological engineering and drilling (Exploration Technology and Engineering).

2.2 Size: researchers, students and graduates (university, official 'Facts and Figures')

Item	Figure / detail	Source
Full-time students	33,125 total: 18,719 undergraduates, 12,208 master's, 2,580 doctoral; 1,035 international	en.cug.edu.cn Facts and Figures
Annual intake	4,633 undergraduates, 4,161 master's, 660 doctoral	same
Graduates per year	4,481 undergraduates, 3,621 master's, 376 doctoral	same
Faculty and staff	3,303 total; 555 professors, 984 associate professors, 714 doctoral supervisors	same
Programmes	70 undergraduate majors; 19 first-level doctoral disciplines; 34 master's disciplines; 16 post-doctoral stations	same
Research facilities	3 State Key Laboratories; 58 provincial/ministerial key labs and centres	same
Faculty of Engineering (closest unit)	214 staff, 145 full-time instructors (48 professors, 74 associate); 1,939 undergraduates and 1,199 graduate students; 5 departments, 1 lab centre, 9 research bases; Geological Engineering rated A+	engineering.cug.edu.cn About

2.3 Academic system

Two-level system: university -> school (xueyuan) -> department / teaching-research section. Undergraduate programmes are 4 years (8 semesters plus summer field terms), ending with a graduation design or thesis and a Bachelor of Engineering or Science. Field teaching is central: all geoscience students attend the Beidaihe (recognition) and Zhoukoudian or Zigui (mapping) field schools, then a production internship at a mine or oilfield.



Selected schools only (CUG lists about 28 schools/colleges). Teal = closest match. Drawn for this report from CUG's official lists of schools and programmes.

2.4 Curriculum

CUG's School of Earth Resources publishes its 2023 training plans (Resource Exploration Engineering 081403; Petroleum Engineering 081502; Resource and Environment Big Data Engineering 081407T) as PDF attachments, but the site blocks automated retrieval, so the semester tables could not be extracted. Because CUG (Wuhan) and CUG (Beijing) share the same origin and the national accreditation standard for this major, the complete Resources Prospecting Engineering curriculum of CUG (Beijing) in Section 3 is representative of both.

Programmes (courses) offered by the relevant schools:

School	Undergraduate programmes
Earth Resources	Resource Exploration Engineering (incl. 'base class', 'big-data', 'new-energy', 'excellent engineer' and Zijin Mining classes); Petroleum Engineering; Resource and Environment Big Data Engineering
Engineering	Geological Engineering; Civil Engineering; Urban Underground Space Engineering; Exploration Technology and Engineering (drilling); Safety Engineering
Geophysics and Geomatics	Exploration Technology and Engineering (Exploration Geophysics); Geophysics; Geological Information Science and Technology
Others relevant	Surveying and Mapping Eng.; Remote Sensing Science and Technology; Materials Science and Engineering; Gemology; Hydrology and Water Resources; Marine Science

Distinctive practical teaching (School of Earth Resources)

- Field and practice bases: Zigui, Xianning-Tongshan, Jianghan Oilfield, Daye (iron mine), Beidaihe and Zhoukoudian.
- Solid Mineral Exploration Experimental Teaching Centre and a Virtual Simulation Teaching Centre for mineral-resource formation, exploration and development.
- 'Ore Deposit Geology Field Training' course (from 2022): students of the national engineering base class rotate through representative deposits - e.g. Pulang copper (Yunnan), Shuiyindong Carlin-type gold (Guizhou), Jiaojia gold (Jiaodong), Dongshengmiao lead-zinc (Inner Mongolia), Daye iron and Tonglushan copper-gold (Hubei), Qarhan potash (Qinghai).

2.5 Laboratories

Laboratory	What it is	Photos (official page)
State Key Laboratory of Geological Processes and Mineral Resources (GPMR)	National lab supporting the location of strategic mineral resources and sustainable resource development	en.cug.edu.cn/info/1195/1459.htm
State Key Laboratory of Biogeology and Environmental Geology (BGEG)	Est. 2011; about 75 research staff; life-environment interaction	en.cug.edu.cn/info/1195/1460.htm
National Engineering Research Center of GIS (NERCGIS)	GIS technology R and D and transfer (MapGIS platform)	en.cug.edu.cn/info/1195/1461.htm
Faculty of Engineering laboratory centre	Instruments for geotechnical, drilling and civil testing; 9 research bases	engineering.cug.edu.cn/Labs/List_of_Labs.htm
Mech. Eng. school - drilling equipment centres	National Engineering Practice and Education Centre with China Geological Equipment Co.; Engineering Research Centre of Rock-soil Drilling and Excavation and Equipment	eniec.cug.edu.cn/info/1008/1046.htm
Yifu Museum (Shaw Museum)	National 4A geological museum on campus used for teaching	cug.edu.cn

Sources for this section:

Facts and Figures: en.cug.edu.cn/About/Facts_and_Figures.htm; List of schools: en.cug.edu.cn/Schools/List_of_schools.htm; Labs: en.cug.edu.cn/Research/Labs___Centres.htm
Faculty of Engineering: engineering.cug.edu.cn/About/About.htm; programme list: eniec.cug.edu.cn/info/1012/1198.htm
School of Earth Resources training plans and field bases: zyxy.cug.edu.cn (pages 'pyfa', info/1030/19849, info/1093/6956)

3. China University of Geosciences (Beijing) - CUGB

3.1 Comparable units

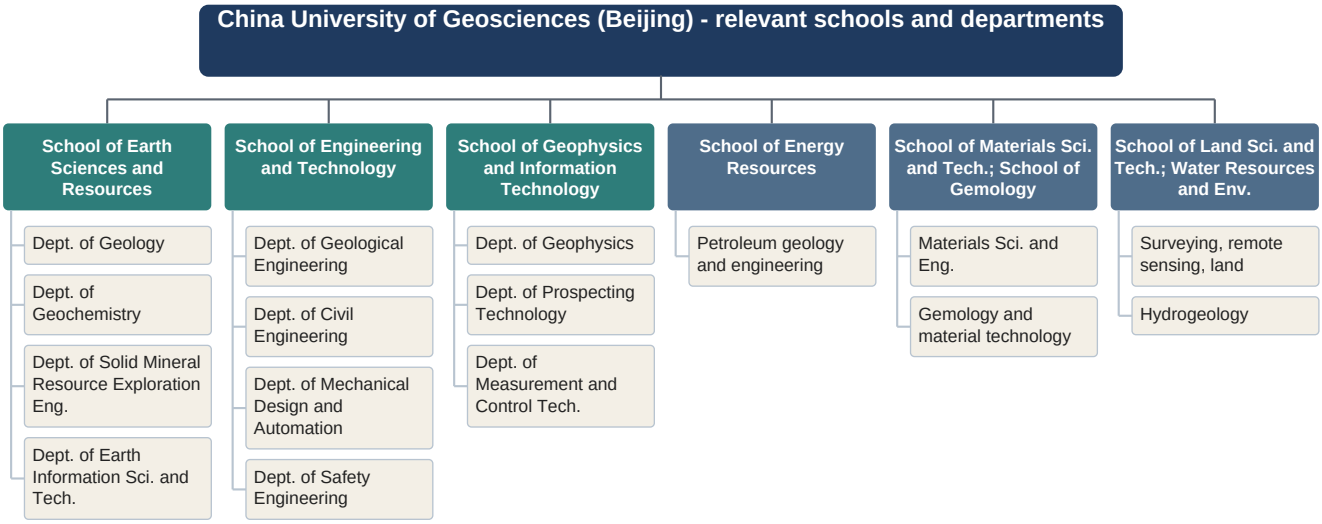
CUGB is the Beijing sister institution of CUG (legally separate since 2005). Its School of Earth Sciences and Resources (geology, geochemistry, solid mineral exploration) and School of Engineering and Technology (geological engineering, civil engineering, drilling machinery, safety) match your Institutes of Resource Exploration and Civil Engineering plus the Drill Engineering Division; the School of Geophysics and Information Technology matches your Institute of Geophysical Prospecting, including its equipment (measurement and control) specialisation.

3.2 Size

Item	Figure / detail	Source
Students	16,758 full-time-equivalent students; 41:59 female:male; about 1% international	Times Higher Education data
Staff ratio	16.6 students per staff member	Times Higher Education
Senior researchers	11 academicians of the Chinese Academy of Sciences and 1 of the Chinese Academy of Engineering	Public encyclopaedia summary
Schools	17 schools offering undergraduate, graduate and postdoctoral programmes	Times Higher Education

CUGB does not publish department-level student or graduate counts in English; use the CUG (Wuhan) figures for scale comparison.

3.3 Academic system



Drawn for this report from CUGB's published list of schools and departments. Teal = closest match.

3.4 Curriculum: Resources Prospecting Engineering (solid minerals)

Objective: graduates able to work as resource-exploration geological engineers - mineral exploration, evaluation and management - and, after about 5 years of practice, to lead technical work in production or research units. Four-year programme, Bachelor of Engineering, built to the national engineering-education accreditation (12 graduate attributes).

Course category	Hours	Credits
General education - required	624	39
Disciplinary fundamental courses	912	57
Specialised core courses	432-448	27-28
Practice teaching (field schools, internships, thesis)	512 / 34 weeks	32
General education - electives	256	16
Creativity and entrepreneurship practice	96	6

Course category	Hours	Credits
Minimum total for graduation		177

Core courses: Ore Deposit Geology; Geochemistry; Remote Sensing Geology; Mineral Prospecting and Exploration; Ore Petrology and Microscopy; Exploration Geochemistry; Introduction to Mining and Mineral Processing. Distinctive practice: Beidaihe and Zhoukoudian field trips, Mineral Exploration course design, integrated training at a mine base, production field trip.

3.5 Subjects by semester with credits (lecture / lab hours in summary)

*Semester numbers are those in the official plan (1-8; 'summer' = summer term after that semester). Content summaries marked * were written for this report.*

Semester	Code	Subject	Cr.	Content summary
Sem 1	GE	Ethics and Fundamentals of Law	3	Required political course
	GE	College English (1-2)	6	General English
	GE	Fundamentals of Computer Science	2	Computer literacy*
	GE	Mental Health; PE (1); Freshman seminar; Intro to the profession	1+1+1+1	University requirements
	DF	Advanced Mathematics A1	6	96 h; single-variable calculus*
	DF	College Chemistry	3	48 h; general chemistry*
Sem 2	GE	College English (3-4)	6	General English
	GE	Computer Language Programming	4	Programming (C/Python)*
	GE	PE (2); Career Planning (1)	1+1	University requirements
	DF	Advanced Mathematics A2	6	Multivariable calculus, series*
	DF	College Physics (1)	3	Mechanics, heat*
	DF	Geosciences (Intro to Earth Science)	4	32 lec + 32 lab; earth materials, processes, plate tectonics*
	PR	Chemistry Experiments	2	48 h lab
Summer 1	PR	Military Theory and Training	2	2 weeks
Sem 3	DF	Linear Algebra	2	Matrices, linear systems*
	DF	Probability and Mathematical Statistics A	3	Statistics for geodata*
	DF	College Physics (2)	3	Electromagnetism, optics*
	DF	Physical Chemistry C	3	Thermodynamics, phase equilibria*
	DF	Crystallography and Mineralogy	5	40 lec + 40 lab; crystal symmetry, mineral classes, hand-specimen ID*
	DF	Crystal Optics	2	16 lec + 16 lab; polarising microscope*
	PR	Physics Experiments	2	48 h lab
	GE	Modern Chinese History; PE (3)	2+1	University requirements
Sem 4	DF	Surveying A	2.5	24 lec + 16 lab; levelling, total station, GNSS*
	DF	Paleontology	2.5	Fossil groups and biostratigraphy*
	DF	Historical Geology	2.5	Stratigraphy and Earth history*
	DF	Petrology	5.5	44 lec + 44 lab; igneous, sedimentary, metamorphic rocks*
	DF	Structural Geology	4	32 lec + 32 lab; folds, faults, stress-strain, maps*
	GE	Principles of Marxism; PE (4)	3+1	University requirements
Summer 2/4	PR	Geological Field Trip, Beidaihe	2	2 weeks; recognition field school
	PR	Geological Mapping Field School, Zhoukoudian	6	6 weeks; regional mapping

Semester	Code	Subject	Cr.	Content summary
Sem 5	SC	Ore Deposit Geology	4	44 lec + 20 lab; ore-forming processes and deposit types*
	SC	Geochemistry (incl. organic)	3	Element distribution, isotopes, organic geochemistry*
	SC	Remote Sensing Geology	3	Image interpretation for mapping and alteration*
	SC	Ore Petrology and Microscopy	3	16 lec + 32 lab; reflected-light ore microscopy*
	SC	Exploration Geophysics Methods and Applications	2	Gravity, magnetic, electrical, EM methods*
Sem 6	SC	Exploration Geochemistry	2	Soil/stream/rock surveys, anomalies*
	SC	Mineral Prospecting and Exploration	3	Exploration stages, drilling grids, resource estimation*
	SC	Mineral Resources Economics and Management	2	Valuation, mineral policy*
	PR	Mineral Exploration Course Design	2	2 weeks; design of an exploration programme
	PR	Political Social Practice	2	2 weeks
	GE	Intro to Mao Zedong Thought...; Career Planning (2)	4+1	University requirements
Summer 6	PR	Integrated Resource-Exploration Training at a Mine Base	2	2 weeks
	PR	Production Field Trip (Resources Prospecting)	6	6 weeks at a mine / exploration team
Sem 7	SC	Professional English for Resource Prospecting	2	Technical reading and writing*
	SC	Frontiers of the Discipline	1	Seminar series
	SC	Introduction to Mining and Mineral Processing	1	Mining methods, beneficiation*
	SC	Environmental Law and Mineral Resources Law	1	Legal framework*
	SC	Solid Mineral Exploration Standards	1	National exploration codes and reporting*
Sem 8	PR	Graduation Design / BS Thesis	6	12 weeks
	GE	Situation and Policies	2	University requirement

Codes: GE general education, DF disciplinary fundamental, SC specialised core, PR practice. Not semester-bound: humanities/social science electives 6, natural science electives 6, entrepreneurship 2, social practice 2, research practice 2, innovation activity 2 credits.

3.6 Laboratories

Laboratory / base	What it is	Photos (official page)
Laboratory of Deep Geological Drilling Technology	Key laboratory of the Ministry of Natural Resources - directly comparable to your Drill Engineering Division	cugb.edu.cn (School of Engineering and Technology)
Cooperation Base for Deep Drilling Equipment and Machinery	Key platform of the Ministry of Science and Technology	cugb.edu.cn
Beidaihe and Zhoukoudian field teaching bases	University-owned field stations used for the 2-week and 6-week field schools	sesr.cugb.edu.cn
Mineral exploration and ore microscopy teaching labs	Support Ore Petrology and Microscopy and Mineral Exploration courses	sesr.cugb.edu.cn

Sources for this section:

Resources Prospecting Engineering programme: sesr.cugb.edu.cn/c/2020/06/17/619374.shtml
Schools, departments, academicians, key labs: en.wikipedia.org/wiki/China_University_of_Geosciences_(Beijing)
Student statistics: timeshighereducation.com/world-university-rankings/china-university-geosciences-beijing

4. Massachusetts Institute of Technology (MIT, USA)

4.1 Comparable unit

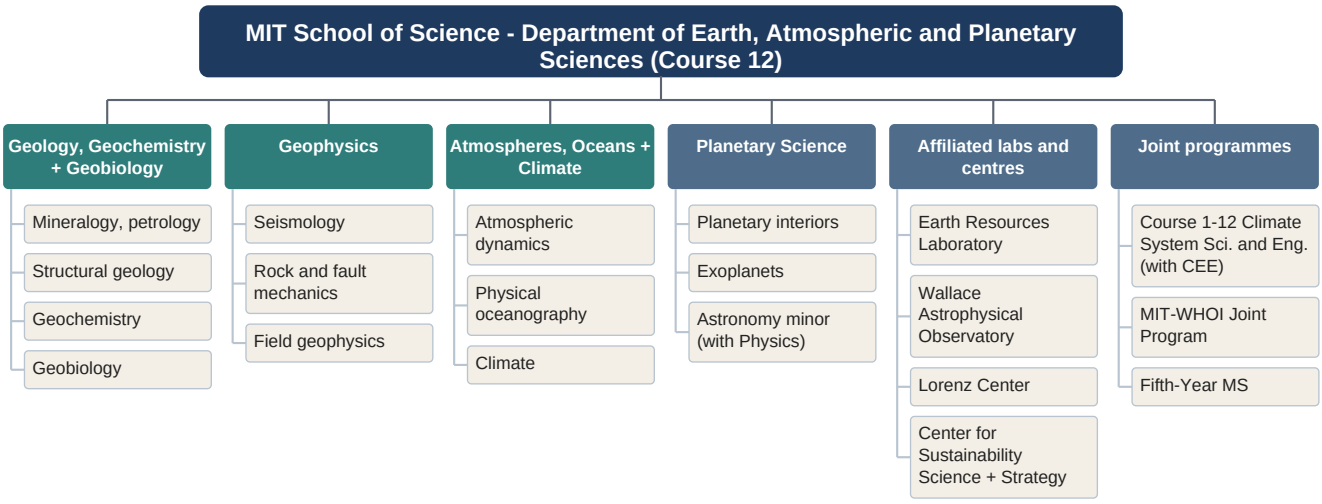
The Department of Earth, Atmospheric and Planetary Sciences (EAPS, 'Course 12') in the School of Science. Geology and Mining Engineering was one of MIT's six original courses (1861); in 1983 the Department of Meteorology and Physical Oceanography merged with Earth and Planetary Sciences to form EAPS. EAPS covers geology, geochemistry, geophysics, atmosphere/ocean/climate and planetary science - i.e. your Institute of Resource Exploration, Institute of Geophysical Prospecting and Laboratory of Space. Civil engineering is in a separate department (Civil and Environmental Engineering, Course 1), with which EAPS runs the joint 'Course 1-12' degree. MIT has no mining, drilling or petroleum programme.

4.2 Size

Item	Figure / detail	Source
EAPS faculty	40+ faculty; includes 3 MacArthur Fellows, 20 members of the National Academy of Sciences, 24 of the American Academy of Arts and Sciences	eaps.mit.edu/about (2026)
EAPS researchers in training	60 postdoctoral scholars	same
EAPS students	175 graduate students; 25 undergraduate majors	same
MIT as a whole (2023-24)	11,920 students (4,576 undergraduate, 7,344 graduate); 1,089 faculty	MIT Facts 2024

4.3 Academic system

MIT undergraduates take the General Institute Requirements (GIRs: calculus, physics, chemistry, biology, humanities, lab, communication) mostly in Year 1 and declare a major at the end of the first spring term. The Course 12 major is flexible: a core of introductory subjects, laboratory/field and computation requirements, one of four concentrations, supporting science subjects and a thesis. Graduate study is organised in four programmes plus the MIT-Woods Hole Oceanographic Institution Joint Program.



Drawn for this report from EAPS's published graduate programmes and affiliated labs.

4.4 Curriculum totals

Requirement	Units	Notes
General Institute Requirements (GIRs)	~17 subjects	Calculus, physics, chemistry, biology, humanities/arts/social sciences, lab, REST, communication
Course 12 major (Earth Science track)	144-150	Breakdown below
Major + unrestricted electives beyond GIRs	180-198	Total for degree beyond GIRs
Conversion	12 MIT units = about 4 US semester credit hours	A standard subject is 12 units

4.5 Course 12 structure and subjects - Earth Science concentration (2026)

Block	Units	Subjects to choose from (12 units each unless noted)
Introductory core	36	Choose from 12.001 Introduction to Geology; 12.002 Introduction to Geophysics and Planetary Science; 12.003 Introduction to Atmosphere, Ocean and Climate Dynamics; 12.004 Introduction to the Global Climate Cycle
Laboratory and field	12-15	e.g. 12.115 Field Geology + 12.116 Analysis of Geologic Data; 12.307 Weather and Climate Laboratory; 12.335 Experimental Atmospheric Chemistry; 12.410J Observational Techniques of Optical Astronomy
Computation	12	12.010 Computational Methods of Scientific Programming; 12.012 MATLAB, Statistics, Regression, Signal Processing; 12.C25J Real World Computation with Julia; 6.100A/B Python and data science; 16.C20J
Earth Science concentration	36-39	12.100 Plate Tectonics and Climate; 12.104 Geochemistry of Natural Waters; 12.107 Geobiology; 12.108 Earth Materials: Minerals and Rocks; 12.110A+B Sedimentary Environments / Field Sedimentology; 12.113 Structural Geology; 12.163 Geomorphology; 12.201 Essentials of Global Geophysics; 12.202 Flow, Deformation and Fracture; 12.203 Mechanics of Earth; 12.210 Introduction to Seismology; 12.211 Field Geophysics; 12.225 Mechanisms of Faulting and Earthquakes; 12.421 Remote Sensing
Supporting subjects	36	e.g. 5.601+5.602 Thermodynamics; 7.05 Biochemistry; 18.03 Differential Equations or 18.06 Linear Algebra
Thesis	6 + at least 6	12.TIP Thesis Preparation; 12.THU Undergraduate Thesis

4.6 Illustrative 8-semester sequence (Earth Science track)

MIT does not fix subjects to semesters; the department publishes a 'roadmap'. The sequence below is an illustrative arrangement of the requirements above, prepared for this report.

Semester	Code	Subject	Cr.	Content summary
Y1 Fall	GIR	Calculus I (18.01), Physics I (8.01), Chemistry (5.111), HASS	48 units	Institute requirements
Y1 Spring	GIR	Calculus II (18.02), Physics II (8.02), Biology (7.01x), HASS	48 units	Institute requirements
Y2 Fall	12.001	Introduction to Geology	12	Earth materials, plate tectonics, field trips
	12.010	Computational Methods of Scientific Programming	12	Programming for scientific problems
	18.03	Differential Equations	12	Supporting maths
Y2 Spring	12.002	Intro to Geophysics and Planetary Science	12	Gravity, magnetism, seismology, planetary interiors
	12.108	Earth Materials: Minerals and Rocks	12	Mineralogy and petrology with lab
Y3 Fall	12.113	Structural Geology	12	Deformation, folds, faults, stress analysis
	12.201	Essentials of Global Geophysics	12	Physics of Earth's interior
Y3 Spring	12.115/116	Field Geology + Analysis of Geologic Data	12-15	Field mapping course and data analysis
	12.210	Introduction to Seismology	12	Seismic waves and sources
Y4 Fall	12.TIP	Thesis Preparation	6	Research proposal
	12.421	Remote Sensing: Physical Principles and Techniques	12	Concentration elective
Y4 Spring	12.THU	Undergraduate Thesis	6+	Independent research thesis

4.7 Laboratories

Laboratory	What it is	Photos (official page)
Earth Resources Laboratory (ERL)	Industry-linked research on subsurface imaging, rock physics, geophysical exploration and energy resources - the MIT unit closest to your geophysical and petroleum institutes	erlweb.mit.edu
Wallace Astrophysical Observatory	Teaching and research observatory (astronomy, planetary science)	wallace.mit.edu
Lorenz Center	Climate science research centre	lorenz.mit.edu
Center for Sustainability Science and Strategy	Earth-system and policy modelling	cs3.mit.edu

Laboratory	What it is	Photos (official page)
MIT-WHOI Joint Program	Ocean science and engineering with Woods Hole Oceanographic Institution	eaps.mit.edu/education/graduate-program/mit-whoi-joint-program

Sources for this section:

EAPS About: eaps.mit.edu/about; Major programmes: eaps.mit.edu/education/undergraduate-program/major-programs
2026 Earth Science degree track: eaps.mit.edu/wp-content/uploads/2026/04/2026-earth-science-degree-track.pdf
MIT Bulletin, EAPS: catalog.mit.edu/schools/science/earth-atmospheric-planetary-sciences; MIT Facts 2024: facts.mit.edu

5. Colorado School of Mines (Golden, Colorado, USA)

5.1 Comparable unit

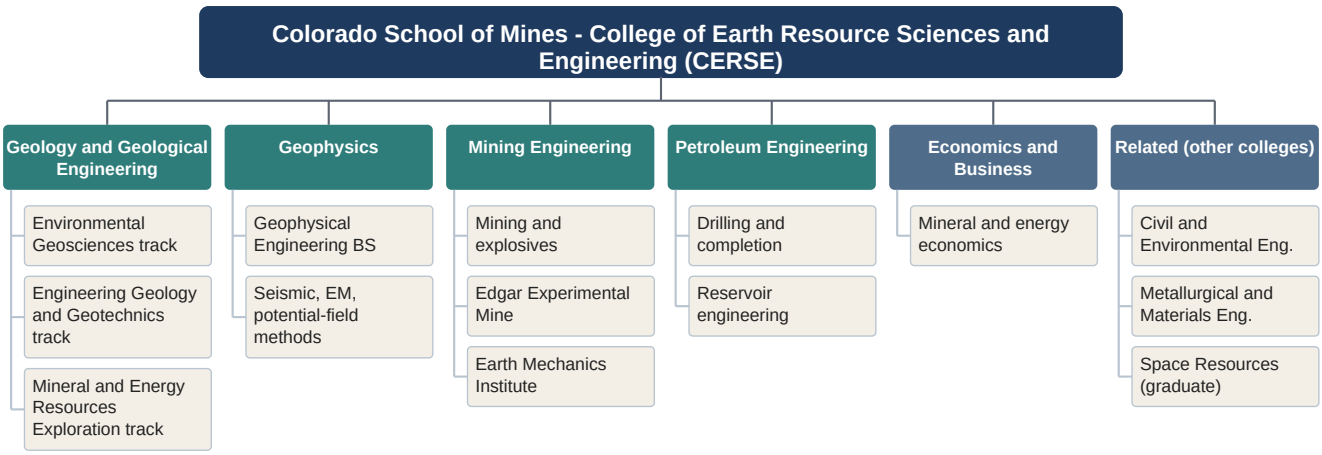
The College of Earth Resource Sciences and Engineering (CERSE) - the school's original, foundational programmes, focused on resource discovery and recovery: Geology and Geological Engineering, Geophysics, Mining Engineering, Petroleum Engineering, and Economics and Business. Civil and Environmental Engineering sits in the College of Engineering and Computational Sciences. Mines is described as the only institution offering doctoral programmes in all five major earth-resource fields (geology and geological engineering, geophysics, geochemistry, mining and petroleum engineering). Of the five universities, Mines is the best Western match to your Institutes of Resource Exploration, Geophysical Prospecting, Petroleum and Mineral Engineering.

5.2 Size

Item	Figure / detail	Source
Students (fall 2025)	8,247 total: 6,371 undergraduate, 1,876 graduate	Wikipedia / mines.edu rankings page
Academic staff	603	Wikipedia infobox
Geology and Geological Eng. faculty	13 professors (incl. interim head), 1 associate, 2 assistant, 5 research professors, 4 research assistant professors, 1 research associate professor, 2 teaching faculty (2025-26 catalogue listing)	catalog.mines.edu
Degrees awarded (2024)	1,997 across all programmes	Data USA
Research classification	R1 - very high research activity	Wikipedia

5.3 Academic system

Semester system (fall, spring, two summer sessions). Year 1 is a common core (calculus, chemistry, physics, design, computing, humanities). Geological Engineering students follow a common Years 1-2, choose one of three tracks after the sophomore year, attend a 5-week summer field camp after Year 3, and finish with capstone design courses in Year 4. The BS is ABET-accredited and prepares students for the Fundamentals of Engineering and Fundamentals of Geology exams.



Drawn for this report from Mines catalogue and college descriptions. Teal = closest match.

5.4 Curriculum: BS Geological Engineering

Item	Detail
Total	130 semester hours (all three tracks)
Common years	Year 1: 31 h; Year 2: 34 h - identical for all tracks
Field session	GEGN316 Field Geology, 5 semester hours, 5 weeks in Colorado and Utah after Year 3
Tracks	Environmental Geosciences; Engineering Geology and Geotechnics; Mineral and Energy Resources Exploration
Other CERSE BS degrees	Geophysical Engineering; Mining Engineering; Petroleum Engineering; Economics

5.5 Subjects by semester - Mineral and Energy Resources Exploration track

Course content paraphrased from the official Mines catalogue descriptions.

Semester	Code	Subject	Cr.	Content summary
Year 1 (Fall + Spring)	GEGN101	Earth and Environmental Systems	4	Lithosphere, hydrosphere, atmosphere, biosphere; resources, hazards; 3 h lecture + 3 h lab
	MATH111	Calculus for Scientists and Engineers I	4	Single-variable calculus
	MATH112	Calculus for Scientists and Engineers II	4	Integration techniques, series
	CHGN121	Principles of Chemistry I	4	General chemistry with lab
	PHGN100	Physics I - Mechanics	4	Classical mechanics with lab
	EDNS151	Cornerstone - Design I	3	First-year engineering design projects
	CSCI128	Computer Science for STEM	3	Programming for engineers
	HASS100	Nature and Human Values	3	Humanities core
	CSM101 / S&W	Freshman Success Seminar; Success and Wellness	1+1	Orientation
Y2 Fall	GEGN203	Engineering Terrain Analysis	3	Geomorphology, landforms and processes; geomorphic maps; Colorado fieldwork
	CHGN122	Principles of Chemistry II	4	Equilibria, kinetics, thermochemistry
	CEEN241	Statics	3	Forces, equilibrium, trusses
	MATH213	Calculus III	4	Multivariable calculus
	HASS215	Futures	3	Humanities core
Y2 Spring	PHGN200	Physics II - Electromagnetism and Optics	4	Fields, circuits, optics
	CEEN311	Mechanics of Materials	3	Stress, strain, beams
	GEGN212	The Rock Cycle	4	Mineralogy and petrology foundations; mineral and rock identification in hand sample
	GEGN217	Geologic Field Methods	2	Field description, topographic maps, structural data, cross-sections; weekend field exercises
	CSM202	Student Well-being	1	-
	CAS	Culture and Society elective	3	-
Y3 Fall	GEOL309	Structural Geology and Tectonics	4	Stress/strain, rock mechanics, mountain belts, rifts; stereonets, seismic sections
	GEOL321	Mineralogy and Mineral Characterization	3	Crystal chemistry; XRD, SEM and optical microscopy labs
	MATH225	Differential Equations	3	ODEs for engineers
	EBGN321	Engineering Economics	3	Cash flow, investment appraisal
Y3 Spring	GEGN432	Geological Data Management	3	Statistics, sampling, regression, programming with geological data sets
	GEGN307	Petrology	4	Igneous, sedimentary, metamorphic processes; megascopic and microscopic study
	GEOL314	Stratigraphy	4	Facies, sequences, basin analysis; lab and field
	GEGN317	Geologic Field Skills	1	Advanced field methods; preparation and safety for field camp
	CAS	Culture and Society elective	3	-

Semester	Code	Subject	Cr.	Content summary
Y3 Summer	GEGN316	Field Geology	5	5 weeks mapping igneous, metamorphic and sedimentary terrain; mining, energy and engineering problems; formal reports
Y4 Fall	GEGN401	Mineral Deposits	4	Magmatic, hydrothermal and sedimentary ore deposits; ore-forming processes; exploration sequences; hand-specimen labs
	GEGN438	Petroleum Geology	4	Source, reservoir, traps, migration; well-log analysis, correlation, exploration exercises
	CEEN312 / MNGN321	Soil Mechanics or Intro to Rock Mechanics	3	Restricted elective
	CAS / FREE	Culture and Society 400-level; free elective	3+3	-
Y4 Spring	GEGN403	Mineral Exploration Design	3	Capstone: commodity and target selection, genetic models, property acquisition, reserve estimate, preliminary economics
	GEGN439	Petroleum Exploration Design	3	Capstone: integrated well and 3D seismic interpretation, prospect generation (Petrel)
	GEGN475	Applications of GIS	3	GIS data structures, DEMs, spatial modelling
	Restricted / FREE	Restricted elective; free elective	3+3	-

Track differences (Year 3-4): Engineering Geology and Geotechnics takes Rock Mechanics, Soil Mechanics + lab, Geological Fluid Mechanics, Engineering Geology and Geotechnics (4), Groundwater Engineering (3+1), Site Investigation (drilling and sampling methods), Engineering Geology Design and Groundwater Engineering Design. Environmental Geosciences takes Geochemistry, Groundwater Engineering and Groundwater Engineering Design (well fields, drilling programmes, pump tests).

5.6 Laboratories

Laboratory	What it is	Photos (official page)
Edgar Experimental Mine (Idaho Springs)	Former 1870s silver-gold mine owned by Mines since 1921; underground lab for mine surveying, geological mapping, blasting, ventilation, rock-mechanics instrumentation and mine safety; also hosts an underground physics institute	mining.mines.edu/edgar-experimental-mine; tour.mines.edu/edgar-mine
Excavation Engineering and Earth Mechanics Institute (EMI)	Est. 1974; excavation technology for mining and civil underground construction	mining.mines.edu/research
Kroll Institute for Extractive Metallurgy	Mineral processing and extractive metallurgy research	mining.mines.edu/research
Explosives engineering lab (near Idaho Springs)	Hands-on blasting and explosives training for mining and related industries	minesnewsroom.com
Mines Museum of Earth Science	Mineral and rock collections from Colorado mining districts and worldwide; used in teaching labs	mines.edu (Museum)
Center for Mineral Resources Science; Center for Advanced Subsurface Earth Resource Models	Department research centres for ore deposits and subsurface modelling	geology.mines.edu

Sources for this section:

BS Geological Engineering (2025-26 catalogue): catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/
Colleges and CERSE description: keenweb (Mines institutional profile); Britannica, 'Colorado School of Mines'
Enrolment: mines.edu/about/rankings; en.wikipedia.org/wiki/Colorado_School_of_Mines
Edgar Mine and research institutes: mining.mines.edu/edgar-experimental-mine; mining.mines.edu/research

6. Interpretation: what the benchmarks show

- Structure: CUG (Wuhan/Beijing) and Colorado School of Mines organise geoscience by applied engineering field (exploration, geophysics, petroleum, mining, drilling, civil/geotechnical), as your faculty does. Tsinghua and MIT organise by science discipline (earth system, atmosphere, planetary) and leave civil engineering to a separate department.
- Drilling: only CUG has a stand-alone drilling programme (Exploration Technology and Engineering) with a dedicated deep-drilling key laboratory and an equipment centre - a direct model for your independent Drill Engineering Division. At Mines drilling is taught inside Petroleum Engineering and the Geological Engineering site-investigation course.
- Credit load: Chinese 4-year engineering degrees require 163-177 credits (Tsinghua 163-169, CUGB 177); US degrees about 130 semester hours. Chinese plans carry more political/general courses and longer field schools; US plans give more electives.
- Field teaching: every benchmark makes field mapping a required, credit-bearing block - CUGB 8 weeks of field schools plus a 6-week production trip; Mines a 5-week field camp; Tsinghua surveying, geology and construction internships in summer terms; MIT a field geology subject.
- Common spine in Years 1-2: calculus (10-12 credits), physics with lab, chemistry, programming, mechanics, then mineralogy-petrology-structural geology in Year 2 - consistent with your fundamental subjects (maths, physics, chemistry, mechanics, fluids, electrical/electronic engineering, programming).
- Capstones: Mines' Mineral Exploration Design and Petroleum Exploration Design, CUGB's Mineral Exploration Course Design and mine-base integrated training, and Tsinghua's Integrated Course Design are good templates for a final-year design project in each of your specialisations.

Gaps in the published data

- Department-level student and graduate counts are not published by Tsinghua, CUGB or Mines; university-level figures are given instead.
- CUG (Wuhan) training-plan PDFs could not be retrieved automatically; they can be downloaded manually from zyxy.cug.edu.cn (2023 training plans) if a Wuhan-specific semester table is needed.
- Laboratory photographs: see the official pages listed in each laboratory table.